

Cross-Border Export Compliance Auditor

An AI-powered n8n workflow that automatically reads export documents, verifies compliance with international trade regulations, and notifies exporters — with zero manual document review.

PREPARED BY

Group 5 — 29 Participants

CATEGORY

AI Workflow Automation

TOOLS

n8n · OpenAI · Firecrawl

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1. Introduction

Project Title

Cross-Border Export Compliance Auditor

Project Description

The **Cross-Border Export Compliance Auditor** is an intelligent workflow built using **n8n** and AI automation. It automatically retrieves export document submissions sent through Gmail, extracts the attached PDF documents, analyzes their contents using Artificial Intelligence, verifies compliance with export regulations, sends approval or amendment emails to applicants, logs every processed application into Google Sheets, and marks processed emails to prevent duplicate handling.

In short: an exporter emails their shipping documents the same way they would submit paperwork for a visa application. From that point on, the workflow takes over — reading, checking, deciding, and replying — without a human needing to open a single attachment.

The workflow eliminates manual document verification, reduces processing time, improves compliance accuracy, and automates communication with exporters.

2. Project Objectives

The objectives of this project are to:

- Automate export document processing.
- Reduce manual verification effort.
- Detect incomplete or missing submissions.
- Verify regulatory compliance using AI.
- Notify exporters automatically of the outcome.
- Maintain a reliable audit trail.
- Prevent duplicate processing of the same email.

3. Workflow Architecture

The diagram below shows the complete workflow as built in n8n — from the scheduled trigger that checks Gmail, through document extraction and AI analysis, to the final compliance decision, notification, and audit logging.

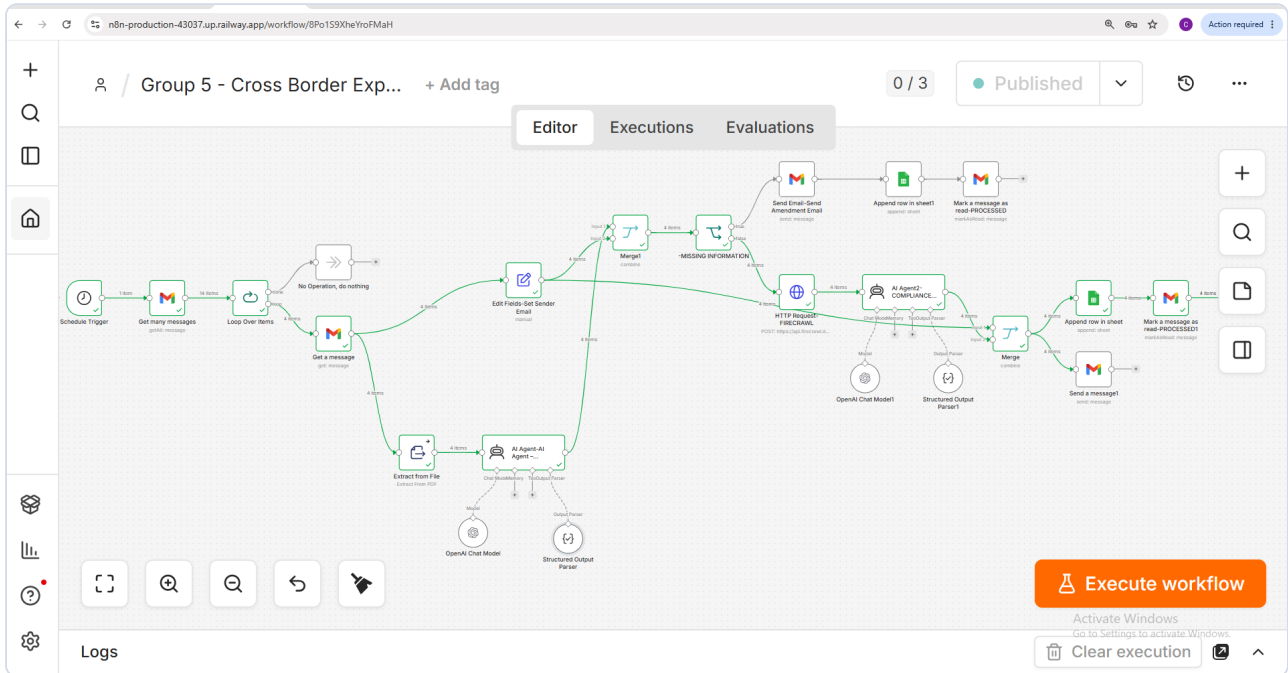


Figure 1 — Full workflow architecture, built in n8n

4. Workflow Explanation

This section breaks down every node in the workflow, what it does, and why it matters — in the order the data actually moves through the system.

4.1 Intake — Receiving the Application

1 Schedule Trigger

Automatically starts the workflow at fixed intervals (every 5 minutes), checking the Gmail inbox for new export applications — no manual execution required.

2 Get Many Messages

Retrieves all unread emails containing attachments, using the Gmail filter `is:unread has:attachment`, ensuring only genuine export submissions are picked up.

3 Loop Over Items

Processes each email one at a time rather than in bulk. This prevents workflow conflicts, simplifies debugging, and guarantees every exporter gets the correct response.

4 Get Message

Downloads the full Gmail message — sender, subject, body, attachments, and Message ID. The Message ID is reused later to mark the email as read.

5 Edit Fields — Set Sender Email

Stores the sender's email and Message ID so they can be reused later when sending responses and marking the email as processed.

4.2 Reading & Understanding the Documents

6 Extract From PDF

Extracts raw text from the PDF attachment (commercial invoice, packing list, certificate of origin, phytosanitary certificate, etc.) so it can be analyzed by AI.

7 AI Agent 1 — Document Extraction

AI

Reads the extracted text and identifies key export details — exporter name, company, importer, destination country, product, HS Code, shipment value, and currency — converting unstructured text into structured JSON.

8 Merge

Combines the sender information (email, Message ID) with the structured export data into a single object for downstream nodes.

What if a document can't be read? If a PDF is damaged, empty, or password protected, the system doesn't stop — it alerts an administrator via Slack so the case gets manual attention, rather than silently failing.

4.3 Validation — Checking for Missing Information

9 Missing Information Check

Verifies that every required field is present — exporter name, destination country, shipment quantity, and more. Missing data routes the application to the amendment path; complete data continues to compliance verification.

10 Send Amendment Email Path A

Automatically emails the applicant a list of missing fields and requests correction — no manual follow-up needed.

→ Continue to Compliance Path B

If the submission is complete, it proceeds directly to regulatory compliance verification.

4.4 Compliance Verification

11 Firecrawl HTTP Request

Retrieves the latest export compliance regulations from the web using Firecrawl, so decisions are always based on current rules rather than outdated static data.

12 AI Agent 2 — Compliance Verification

AI

Compares the applicant's document against the retrieved regulations — checking product eligibility, destination restrictions, required certificates, and HS Code validity — and returns a decision (**Approved** or **Rejected**), reason, violations, and recommendations. This is the same judgment call a customs officer would make manually.

13 Merge

Combines the compliance decision with the applicant's information into one final dataset — applicant details, decision, AI explanation, and recommendations.

Sample AI Compliance Summary

Commodity: Sesame Seeds · Destination: Germany · Certificates: Complete · Restrictions: None · Risk Level: Low

4.5 Decision, Notification & Audit

14 Append Row in Google Sheet

Saves a permanent audit record — timestamp, sender, exporter, company, importer, destination, product, HS Code, shipment value, currency, decision, and reason.

15 Send Approval Email

If approved, the exporter automatically receives a confirmation email with the approval status, AI explanation, recommendations, and a unique reference number (e.g. *EXP-2026-00152*) — no human needs to write it.

16 Mark Email as Read

Uses the Message ID retrieved earlier to mark the email as read, guaranteeing the same application is never processed twice.

5. Artificial Intelligence Components

AI Agent 1 — Document Understanding

- Reads the extracted PDF text
- Extracts exporter, shipment, and product information
- Converts unstructured text into structured data (JSON)

AI Agent 2 — Compliance Verification

- Reads current export regulations (via Firecrawl)
- Compares the applicant's documents against those regulations
- Makes an approval/rejection decision
- Generates supporting reasons and recommendations

6. Error Handling

The workflow is built to handle failure gracefully rather than break silently. It accounts for:

- Missing PDF attachments
- Invalid or corrupted PDF structure
- Missing required export information
- Firecrawl API errors
- Gmail API authentication errors
- Google Sheets API failures

Emails with incomplete documentation are automatically routed to an amendment notification instead of proceeding to compliance analysis — ensuring no incomplete application is ever wrongly approved or silently dropped.

7. Technologies Used

Technology	Purpose
n8n	Workflow automation engine
Gmail API	Retrieve and send emails
OpenAI	AI document extraction and compliance analysis
Firecrawl API	Retrieve up-to-date export regulation content
Google Sheets API	Store audit records
PDF Extract Node	Extract text from PDF documents

8. Workflow Summary

Schedule Trigger

|

▼

Retrieve Unread Emails

|

▼

Loop Through Emails → Get Full Email → Extract PDF

|

▼

AI Agent 1 (Extract Structured Data)

|

▼

Merge Sender Information

|

▼

Missing Information?

└─ Yes → Send Amendment Email

└─ No → Firecrawl → **AI Agent 2** (Compliance Check)

|

▼

Merge Results

└─ Append Row → Google Sheet

└─ Send Approval Email → Mark Email as Read

9. Conclusion

The **Cross-Border Export Compliance Auditor** demonstrates how AI and workflow automation can streamline export document processing end-to-end. By integrating Gmail, PDF extraction, AI-based information extraction, regulatory compliance verification, Google Sheets logging, and automated email notifications, the solution significantly reduces manual effort, improves processing speed, ensures consistent compliance checks, and provides a reliable audit trail for every application submitted.